Noah Krever

Email: ndk2115@columbia.edu **Website:** noahkrever.github.io

RESEARCH INTERESTS

Algorithmic game theory, online multi-agent learning, optimization, AI for social good

EDUCATION

Columbia University | B.A. in Data Science (CS & Stats) with Mathematics concentration May 2022
Cumulative GPA: 3.8/4.0, **Major GPA:** 3.9/4.0

PUBLICATIONS

(P: Preprint & Manuscript, C: Conference & Journal, W: Workshop & Demo, ∗: Equal contribution, †: Equal senior role)

[C1] GUARD: Constructing Realistic 2-Player Matrix & Security Games for Benchmarking Game-Theoretic Algorithms
Noah Krever, Jakub Černý, Moïse Blanchard, Christian Kroer
Conference on Neural Information Processing Systems (NeurIPS) 2025
Spotlight (Top 3% of submissions in D&B track)

[P1] Is Quasar PG1302-102 Periodic?
Noah Krever, Maria Charisi, Daniel D'Orazio, Zoltan Haiman
Manuscript 2022

RESEARCH EXPERIENCE

Columbia IEOR, Lily Xu Group | NY, NY
Researcher, Optimization Oct 2025 – Present

- Optimizing eco-credit allocation mechanisms and decision protocols to maximize Kenyan ecosystem wellbeing
- Modeling ecological metrics & sensor readings for MDPs, multi-armed bandits, and game-theoretic data auditing

Columbia IEOR, Christian Kroer Group | NY, NY
Researcher, Algorithmic Game Theory June 2024 – Present

- Published first-author spotlight paper introducing statistical theorems and our novel game theory framework
- Implemented original suite of standard algorithms for Nash and Stackelberg equilibria, mixed-integer linear programming, incremental strategy generation (double oracle solvers), and self-play with no-regret learners
- Built flexible object-oriented framework for converting real-world data into hyper-realistic security game instances

Columbia University Physics Department, Zoltan Haiman Group | NY, NY
Researcher, Computational Astrophysics Jan. 2019 – May 2022

- Lead project studying false periodicities in quasar time-domain surveys, evaluated binary black-hole candidate
- Utilized HPC cluster for large-scale time-series simulations & machine learning toolkits for statistical analysis
- Scripted algorithms to perform spectral analysis, Markov processes, signal processing, and Bayesian model-fitting

TCNJ Physics Department, Tuan Nguyen Lab | NY, NY
Summer Researcher, Computational Biophysics June 2017 – Aug. 2017

- Mapped network connectivity using single cell laser scanning photo stimulation with calcium imaging
- Wrote original algorithms for automating network analysis, cut processing time from one week to 3 seconds

WORK EXPERIENCE

Roc360 | NY, NY
Data Scientist June 2022 – Present

- Solving open-ended problems in Markov processes, graph theory, network modeling & algorithms
- Training, deploying, automating wide range of machine learning models for classification, ranking, prediction
- Leading development of SQL/Python analytics data layer linking pipelines and models using dbt

Columbia University Computer Science Department | NY, NY
Teaching Assistant, COMS 4701: Artificial Intelligence Jan. 2022 – May 2022

- Led extended office hours teaching AI algorithms and probability/coding concepts, designed review material
- Graded and evaluated numerous projects, written/programming problem sets, and exam submissions

Teaching Assistant, ENGI 1006: Python Sept. 2020 – Dec. 2021

- Led office hours teaching Python fundamentals, designed review material, graded projects, HWs, exams

AWARDS AND HONORS

Spotlight Paper (Top 3%), NeurIPS 2025
GUARD: Constructing Realistic Two-Player Matrix and Security Games for Benchmarking Game-Theoretic Algorithms

Dean's List, Columbia University 2019
Awarded to the Columbia University students maintaining highest GPAs

Kluge Scholar, Columbia University 2018
Scholarship program offered to select students, offers funding, specialized advising, seminars and speaker series

TEACHING EXPERIENCE

Artificial Intelligence (COMS 4771) Spring 2022
Teaching Assistant at Columbia University (Instructor: Prof. Tony Dear)

Intro to Computing in Python (ENGI 1006) Fall 2020, Fall 2021
Teaching Assistant at Columbia University (Instructor: Prof. Daniel Bauer)

SELECTED SIDE PROJECTS

Regret Minimization in 3-Player Poker – Customized CFR+ to solve a simplified variant of post-flop 3-player poker and displayed/verified output strategies. Presented for course IEOR 6998: *AI, Games, and Markets*. **Grade:** 100/100

Understanding Neural ODEs for Time Series – Implemented novel Neural ODE architectures with adjoint sensitivity training for time series forecasting. Presented for course COMS 4995: *Neural Networks & Deep Learning*. **Grade:** 92/100

Traversing Latent Spaces for Interpretability – Novel exploration of deep classifiers' latent spaces, probed decision-boundaries with perturbations & class. loss VAE. Presented for course COMS 6998: *Adv. Deep Learning*. **Grade:** 100/100

ACTIVITIES

Columbia Undergraduate Science Journal (CUSJ), *Editor & Board Member for CU Research Journal* Oct. 2019 – May 2022

Columbia Poker Club, *Founder and President (500+ Member Club Organizer, Teacher, Player)* July 2021 – May 2022

Columbia University Esports Club, *Rocket League Team Captain & Founder (Top 0.1% of players)* June 2020 – May 2022

Columbia University Chess Club, *Member (2200 online ELO)* Sept. 2020 – May 2022